1. What is the difference between an absolute path and a relative path? Give one example of each from your lab.

   Answer

   An **absolute path** specifies the complete location of a file or folder starting from the root drive (such as C:\). It always points to the same location regardless of the current working directory whiles a **relative path** specifies the location of a file or folder relative to the current directory. Its meaning depends on where the command is executed.

   Example of an absolute path from the lab

   C:\Users\Public\CIP-B101_Lab1C_Fuseini_Imoru\outputs\sys_info_report.txt

   Example of a relative path from the lab

   ..\outputs\sys_info_report.txt

2. Why should analysts create a separate lab/evidence folder before running commands?

   Answer

   Analysts should create a separate lab or evidence folder before running commands to ensure that collected data is organized, preserved and protected from accidental modification. Keeping evidence in a dedicated folder helps maintain the integrity of the investigation, prevents important files from being mixed with system files and makes it easier to locate, document and analyze collected information. It also reduces the risk of overwriting or deleting evidence and supports proper chain-of-custody and forensic documentation practices.

3. Which command showed your IP address and which command showed your MAC address?

   Answer

   Command used to display the IP address: ipconfig | findstr IPv4

   Command used to display the MAC address: getmac /v

4. Why is a batch file useful in forensic triage?

   Answer

   A **batch file** is useful in forensic triage because it automates the collection of system information and ensures that the same commands are executed consistently every time. This saves time, reduces human error and helps investigators quickly gather important evidence such as operating system details, hardware information, network configuration and running processes. Batch files also make the evidence collection process repeatable and support proper documentation during incident response and forensic analysis.

5. Why must analysts document failed commands or failed pings instead of hiding them?

   Answer

   Analysts must document failed commands or failed pings because failures are also valuable forensic evidence. A failed command or unreachable host can indicate network restrictions, misconfiguration, system limitations or possible security controls such as firewalls or filtering. In forensic investigations, hiding failures would create an incomplete or misleading picture of the system state. Proper documentation ensures transparency, supports accurate incident reconstruction and maintains the integrity and credibility of the investigation findings.

6. What risk can occur if a suspicious executable is placed in a directory listed in PATH?

   Answer

   Risk

   - **Command hijacking:** A malicious program with a common name (e.g., notepad.exe) could run instead of the legitimate one.
   - **Privilege abuse:** If executed with elevated privileges, the malware may gain high-level access to the system.
   - **Stealth execution:** The user may not realize a malicious program is running because it appears to be a normal command.
   - **System compromise:** Attackers can use this to install malware, steal data, or maintain persistence on the system.